

MINI PROJECT REPORT

Design and PCB Development of a Custom STM32CUBEMX-Based Flight Controller for UAV Applications

Mini Project Report Submitted in Partial fulfilment of the requirements for the degree of Bachelor of Technology

From

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ABSTRACT

A flight controller is the central electronic unit of an Unmanned Aerial Vehicle (UAV), responsible for acquiring sensor data, processing flight information, and controlling the motors to maintain stable flight. The performance and reliability of a UAV largely depend on the design of its flight controller hardware. Therefore, a well-designed schematic and Printed Circuit Board (PCB) are essential for ensuring efficient operation, stable power distribution, and reliable communication among different onboard peripherals.

The objective of this project is to design a custom flight controller hardware based on the STM32CUBEMX microcontroller using Proteus Design Suite. The project focuses exclusively on hardware development, including circuit schematic design, PCB layout generation, and microcontroller pin configuration using STM32CUBEMX. No firmware development or hardware implementation has been carried out as part of this project.

The proposed flight controller integrates essential hardware required for modern UAV systems. These include the ICM-40609-D Inertial Measurement Unit (IMU) for measuring acceleration and angular velocity, the MMC5983MA Magnetometer for heading estimation, and the BMP390L Barometric Pressure Sensor for altitude measurement. The controller also incorporates USB Type-C connectivity, SWD debugging interface, SD card interface, ESC outputs, telemetry connectors, GPS interface, RC receiver interface, RGB status LED, buzzer circuit, and a dedicated power management section consisting of a buck converter, power multiplexer, and low-dropout voltage regulator.

The complete schematic was designed in Proteus by dividing the hardware into multiple functional blocks, including the main controller circuit, sensor interface, power supply, and communication interfaces. After schematic completion, footprints were assigned to all components and the PCB layout was developed following standard PCB design guidelines such as proper component placement, optimized routing, decoupling capacitor placement, and continuous ground plane implementation. The STM32CUBEMX peripheral configuration was also prepared using STM32CUBEMX to allocate communication interfaces including SPI, I²C, UART, USB, SDMMC, PWM, ADC, and GPIO according to the hardware design.

The completed schematic and PCB provide a robust hardware platform that can be used for future firmware development and integration into a custom quadcopter. The modular design allows additional features and software functionality to be implemented in later stages without requiring significant hardware modifications.

CONTENTS

Acknowledgement	2
Abstract	3
Project Members	4
List of Principal Symbols and Acronyms	7
List of Figures	8
List of Tables	9
➤ CHAPTER 1	
➤ INTRODUCTION	10
1.1 Introduction	10
1.2 Literature Review	10
1.3 Objectives	11
1.4 Scope of Present Work	11
➤ CHAPTER 2	
➤ SYSTEM DESIGN AND HARDWARE DESCRIPTION	13
2.1 System Overview	13
2.2 Hardware Architecture	14
2.3 STM32CUBEMX Microcontroller	14
2.4 Sensor Interfaces	15
2.5 Power Management	15
2.6 Communication Interfaces	15
2.7 Working Principle	16
➤ CHAPTER 3	
➤ SCHEMATIC DESIGN, STM32CUBEMX PIN CONFIGURATION AND PCB DESIGN	17
3.1 Introduction	17
3.2 Design Methodology	18
3.3 Main Controller Circuit	19
3.4 Sensor Interface Circuit	19
3.5 Power Management Circuit	20
3.6 USB and External Interface Circuit	20
3.7 STM32CUBEMX Pin Configuration	20
3.8 PCB Design	21
3.9 PCB Layout	22
3.10 PCB Design Considerations	22
➤ CHAPTER 4	
➤ PCB DESIGN VERIFICATION AND ANALYSIS	23

4.1 Introduction	23
4.2 Schematic Verification	23
4.3 STM32CUBEMX Pin Configuration Verification	23
4.4 PCB Layout Analysis	24
4.5 PCB Routing Analysis	24
4.6 Design Rule Verification	25
4.7 Electrical Rule Verification	25
4.8 Advantages of the Proposed Hardware Design	25
4.9 Limitations of the Current Work	26
4.10 Discussion	26
➤ CHAPTER 5	
➤ CONCLUSION AND FUTURE SCOPE	28
5.1 Conclusion	28
5.2 Future Scope	29
5.3 Overall Achievement of the Project	30
REFERENCES	31

LIST OF PRINCIPAL SYMBOLS AND ACRONYMS

Acronym	Full Form
UAV	Unmanned Aerial Vehicle
MCU	Microcontroller Unit
PCB	Printed Circuit Board
IMU	Inertial Measurement Unit
ESC	Electronic Speed Controller
SPI	Serial Peripheral Interface
PC	Inter-Integrated Circuit
UART	Universal Asynchronous Receiver Transmitter
USB	Universal Serial Bus
SDMMC	Secure Digital Multi Media Card
SWD	Serial Wire Debug
PWM	Pulse Width Modulation
ADC	Analog to Digital Converter
GPIO	General Purpose Input Output
RGB	Red Green Blue
LDO	Low Dropout Regulator
GPS	Global Positioning System
RC	Radio Control
SD	Secure Digital
DRC	Design Rule Check
ERC	Electrical Rule Check

LIST OF FIGURES

Figure No.	Title
Figure 1.1	Flowchart of the Scope of the Project
Figure 2.1	Overall Flight Controller Block Diagram
Figure 3.1	Hardware Design Flow
Figure 3.2	Main Controller Schematic (Page 1 of schematic)
Figure 3.3	Sensor Interface Schematic (Page 2 of schematic)
Figure 3.4	Power Supply Schematic (Page 3 of schematic)
Figure 3.5	USB and Interface Connector Schematic (Page 4 of schematic)
Figure 3.6	STM32CUBEMX Pin Configuration
Figure 3.7	PCB Top Layer
Figure 3.8	PCB Bottom Layer
Figure 3.9	3D PCB View
Figure 4.0	PCB Layout
Figure 4.1	Design Rule Check (DRC)
Figure 4.2	Electrical Rule Check (ERC)

LIST OF TABLES

Table No.	Title
Table 2.1	Major Hardware Architecture Blocks
Table 2.2	STM32CUBEMX Specifications
Table 2.3	Sensor Specifications
Table 3.1	Hardware Components Used in Schematic Design
Table 3.2	STM32 Peripheral Configuration
Table 4.1	Advantages of the Proposed Design
Table 5.1	Project Achievement Summary

CHAPTER 1

INTRODUCTION

1.1 Introduction

Unmanned Aerial Vehicles (UAVs), commonly known as drones, have become an important part of modern technology, with applications ranging from aerial photography and surveying to agriculture, logistics, and search-and-rescue operations. At the heart of every UAV lies the flight controller, an embedded electronic system responsible for acquiring data from onboard sensors, processing this information in real time, and generating the control signals required to stabilise and manoeuvre the vehicle. The reliability, responsiveness, and overall performance of a UAV depend heavily on the quality of its flight controller hardware.

Commercially available flight controllers, while functional, often restrict designers to fixed hardware configurations and limited customisation options. Designing a custom flight controller allows a deeper understanding of embedded hardware design, sensor integration, power management, and PCB layout practices, while also providing a platform that can be tailored to specific project requirements. This mini project focuses on the hardware design of a custom flight controller built around the STM32CUBEMX microcontroller, a high-performance ARM Cortex-M7 based device suitable for real-time flight control applications.

The project covers the complete hardware development cycle, from component selection and schematic capture in Proteus Design Suite to PCB layout design and microcontroller peripheral configuration in STM32CUBEMX. The resulting hardware platform is intended to serve as the foundation for future firmware development and integration into a fully functional quadcopter.

1.2 Literature Review

A number of open-source and commercial flight controller platforms are widely used in the UAV community, including Pixhawk-based boards and controllers running the ArduPilot and Betaflight firmware stacks. These platforms typically integrate an IMU, barometer, and magnetometer along with a high-performance microcontroller to perform real-time attitude estimation and control. Studies of these designs highlight several recurring hardware design principles: careful placement of decoupling capacitors near microcontroller power pins, isolation of sensitive sensors from noisy power circuitry, use of continuous ground planes to reduce electromagnetic interference, and modular schematic organisation to simplify debugging and future revisions.

The STM32 family of microcontrollers, particularly the STM32F4 and STM32H7 series, is commonly adopted in flight controller designs due to its high clock speed, rich peripheral set, and floating-point processing capability, all of which are beneficial for sensor fusion algorithms such as the Kalman filter and complementary filter. The STM32CUBEMX in particular offers a

substantial improvement in processing headroom over older STM32F4-based designs, making it a suitable candidate for a modern custom flight controller.

Reviewing existing open-source hardware designs also confirms the importance of standard interfaces such as SWD for programming and debugging, USB for configuration and firmware upload, and SDMMC for onboard data logging. These observations directly informed the hardware architecture adopted in this project.

1.3 Objectives

The objectives of this mini project are as follows:

- To select suitable electronic components for the flight controller design.
- To design the complete schematic using Proteus Design Suite.
- To integrate the STM32CUBEMX microcontroller with essential onboard sensors.
- To design power management and communication interface circuits.
- To configure the STM32CUBEMX peripherals using STM32CUBEMX.
- To develop a compact and reliable PCB layout following standard PCB design guidelines.
- To create a hardware platform that can support future firmware development and UAV implementation.

1.4 Scope of Present Work

The scope of this project is limited to the hardware design and development of a custom flight controller for UAV applications. The work completed in this project includes:

- Selection of the STM32CUBEMX microcontroller.
- Design of the complete flight controller schematic using Proteus.
- Integration of the ICM40609DIMU, MMC5983MA magnetometer, and BMP390L barometer.
- Design of the USB Type-C interface, SWD debugging interface, SD card interface, GPS connector, telemetry connectors, ESC outputs, RGB LED, buzzer, and power management circuits.
- STM32CUBEMX pin configuration for all required peripherals.
- PCB layout design, component placement, routing, and preparation for future fabrication.

The following activities are not included in this mini project:

- Firmware development.
- PCB fabrication.
- Component soldering.
- Hardware testing.
- Sensor calibration.
- Flight testing.

- Autonomous navigation.

These activities are proposed as future work after successful hardware fabrication.

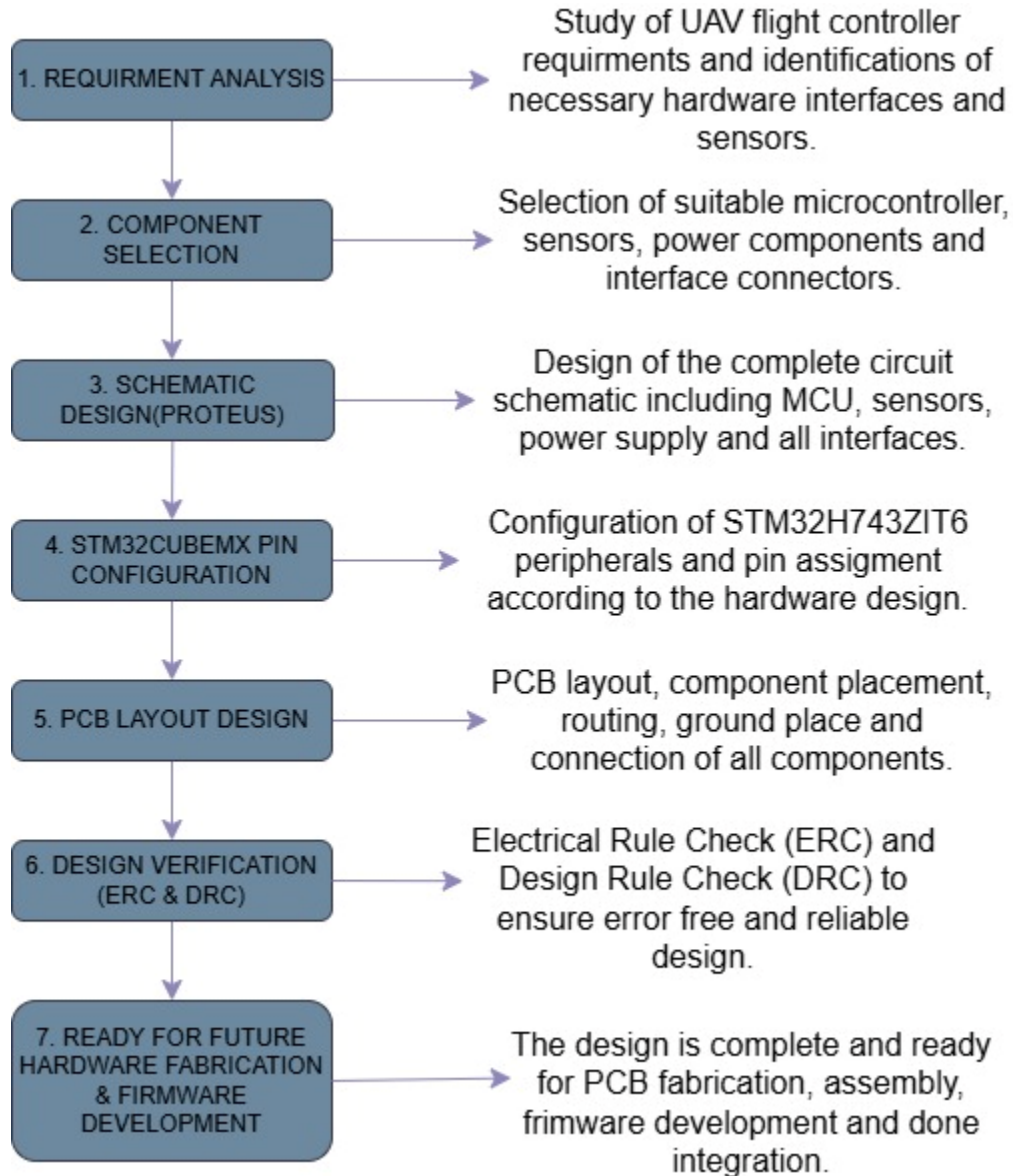


Figure 1.1 -- Flowchart of the Scope of the Project

CHAPTER 2

SYSTEM DESIGN AND HARDWARE DESCRIPTION

2.1 System Overview

The proposed system is a custom-designed flight controller hardware developed for unmanned aerial vehicle (UAV) applications. The design is centred around the STM32CUBEMX microcontroller, which serves as the primary processing unit for all future flight control operations. The objective of this hardware design is to provide a reliable and expandable platform that can later be programmed with flight control firmware and integrated into a quadcopter.

The flight controller schematic has been divided into four functional sections to improve readability, simplify debugging, and support future modifications. These sections are:

- Main Controller Circuit
- Sensor Interface Circuit
- Power Management Circuit
- USB and External Interface Circuit

Each section performs a specific function while working together to form the complete flight controller hardware. The system integrates three essential sensors required for stable UAV operation:

- ICM40609D Inertial Measurement Unit (IMU)
- MMC5983MA Three-axis Magnetometer
- BMP390L Digital Barometric Pressure Sensor

The design also includes communication interfaces such as USB Type-C, GPS, telemetry, RC receiver, SD card interface, ESC outputs, RGB status LED, buzzer, and SWD debugging interface, making the hardware suitable for future firmware implementation.

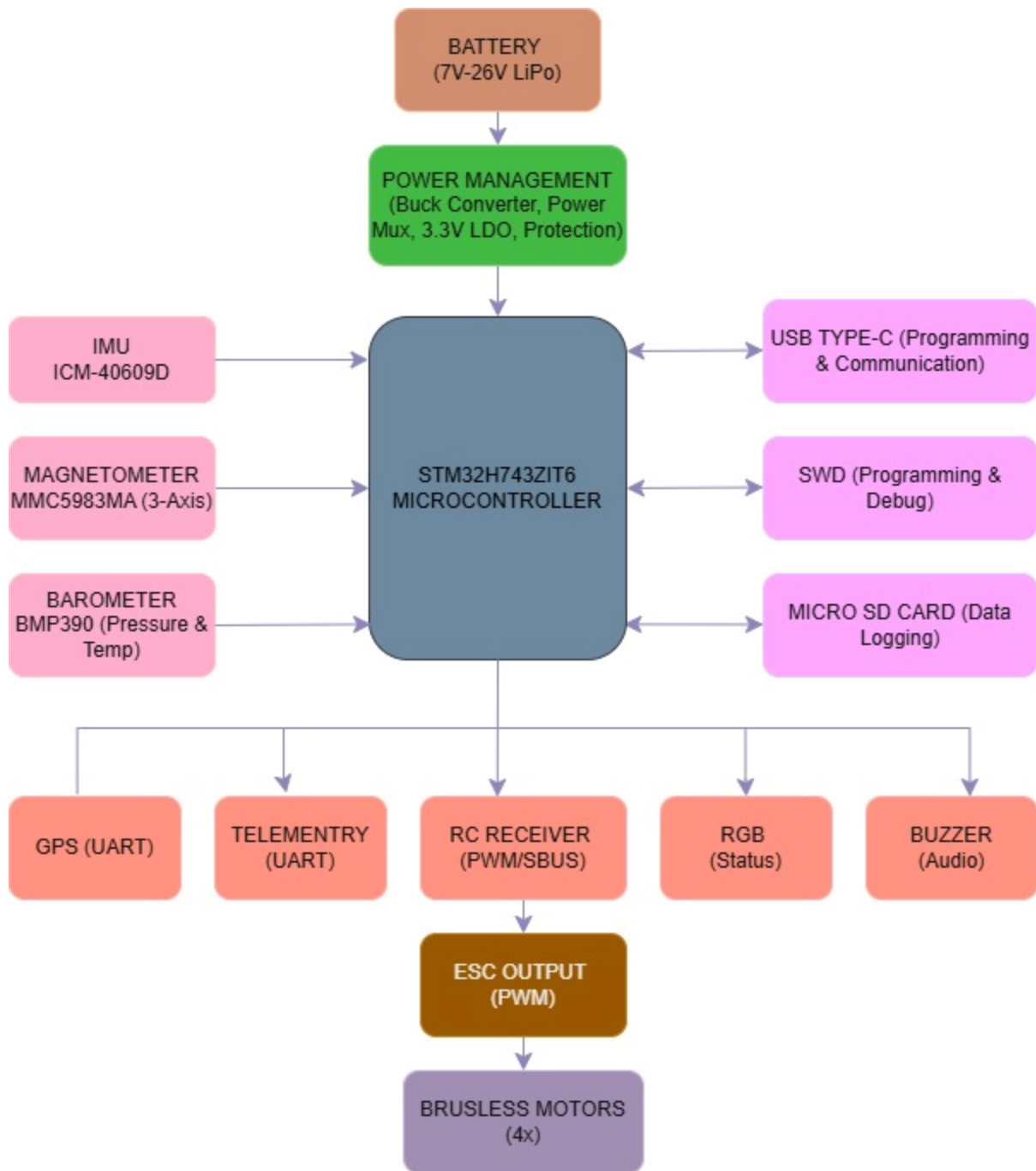


Figure 2.1 -- Overall Flight Controller Block Diagram

Hardware Block	Role in the System
Power Management	Converts and regulates battery input into stable voltage rails for the MCU and sensors

Hardware Block	Role in the System
STM32CUBEMX MCU	Central processing unit; acquires sensor data and will host future flight-control firmware
Sensor Suite	Provides acceleration, angular rate, heading, and altitude measurements
Communication & I/O	Connects the controller to USB, GPS, telemetry, RC receiver, SD card, and ESC outputs

Table 2.1 -- Major Hardware Architecture Blocks

2.2 Hardware Architecture

The proposed flight controller follows a centralised architecture in which all sensors and peripheral devices are directly connected to the STM32CUBEMX microcontroller. The microcontroller communicates with the different peripherals through multiple communication protocols, namely SPI, I²C, UART, USB, and SDMMC. This centralised approach keeps signal routing predictable and simplifies both the schematic layout and the STM32CUBEMX peripheral configuration.

Sensors that require high-speed, low-latency data transfer, such as the IMU and magnetometer, are interfaced through SPI, while the barometer, which requires comparatively lower update rates, is interfaced through I²C. UART channels are reserved for the GPS module and the telemetry radio, USB OTG handles configuration and data transfer through the USB Type-C connector, and SDMMC provides an interface for an onboard microSD card used for future data logging.

2.3 STM32CUBEMX Microcontroller

The STM32CUBEMX is a high-performance microcontroller from the STM32H7 series, built around the ARM Cortex-M7 core. It was selected as the central processing unit of the flight controller because of its high clock speed, large memory capacity, hardware floating-point unit, and extensive set of communication peripherals, all of which are beneficial for real-time sensor fusion and flight control computations in future firmware development.

Parameter	Specification
Core	ARM Cortex-M7 with double-precision FPU
Maximum Clock Speed	480 MHz
Flash Memory	2 MB
SRAM	1 MB
Package	LQFP144
Operating Voltage	1.71 V to 3.6 V

Parameter	Specification
Key Peripherals	SPI, I ² C, UART, USB OTG, SDMMC, ADC, PWM Timers, GPIO

Table 2.2 -- STM32CUBEMX Specifications

2.4 Sensor Interfaces

Three sensors form the core sensing suite of the flight controller, each connected to the STM32CUBEMX through a dedicated communication interface as summarised below.

Sensor	Interface	Measurement
ICM-40609D	SPI	3-axis acceleration and 3-axis angular velocity
MMC5983MA	SPI	3-axis magnetic field for heading estimation
BMP390L	I ² C	Barometric pressure and temperature for altitude estimation

Table 2.3 -- Sensor Specifications

The IMU and magnetometer are positioned away from the power management section on the PCB to reduce electromagnetic interference, while the barometer is shielded from direct airflow to improve the accuracy of altitude readings.

2.5 Power Management

The power management circuit accepts the input battery voltage and converts it into stable, regulated voltage rails for the microcontroller and sensors. The section consists of a buck converter for stepping down the input voltage, a power multiplexer for selecting between multiple possible power sources (such as battery and USB), and a 3.3 V low-dropout (LDO) regulator for generating a clean, low-noise supply for the sensitive sensor and microcontroller circuitry. Power filtering capacitors and protection circuitry are included to improve voltage stability and safeguard the board against transient conditions.

2.6 Communication Interfaces

The flight controller provides a comprehensive set of communication interfaces to support both configuration and future flight operations:

- USB Type-C - firmware upload, configuration, and data transfer.
- SWD - programming and debugging of the STM32CUBEMX.
- SD Card (SDMMC) - onboard data logging.
- GPS Connector (UART) - position and navigation data.

- Telemetry Connector (UART) - ground-station communication.
- RC Receiver Connector - manual pilot control input.
- ESC Outputs (PWM) - motor speed control signals.
- RGB LED and Buzzer (GPIO) - visual and audible status indication.

2.7 Working Principle

During operation, the STM32CUBEMX will continuously read acceleration, angular velocity, magnetic heading, and barometric pressure data from the onboard sensor suite. This raw sensor data can then be processed by future firmware using sensor-fusion algorithms to estimate the orientation and altitude of the UAV. Based on this estimated state and the pilot commands received through the RC receiver or telemetry link, the microcontroller will generate the PWM signals required to drive the Electronic Speed Controllers (ESCs) and, in turn, the brushless motors, thereby maintaining stable flight. The USB, SWD, and SD card interfaces support development, debugging, and data logging throughout this process, while the RGB LED and buzzer provide real-time status feedback.

CHAPTER 3

SCHEMATIC DESIGN, STM32CUBEMX PIN CONFIGURATION AND PCB DESIGN

3.1 Introduction

The hardware design of a flight controller is one of the most critical stages in UAV development. A properly designed schematic and PCB ensure reliable power distribution, stable communication between components, reduced electrical noise, and ease of future firmware development. In this project, the complete flight controller hardware was designed using Proteus Design Suite, while the peripheral configuration of the STM32CUBEMX microcontroller was prepared using STM32CUBEMX.

The development process followed a systematic approach beginning with component selection, followed by schematic capture, pin assignment, PCB layout generation, and design verification. The schematic was divided into functional sections to simplify design and improve readability. After successful completion of the schematic, footprints were assigned to all components, and a PCB layout was generated with careful attention to routing, grounding, and component placement.

The overall hardware design process is illustrated in Figure 3.1.

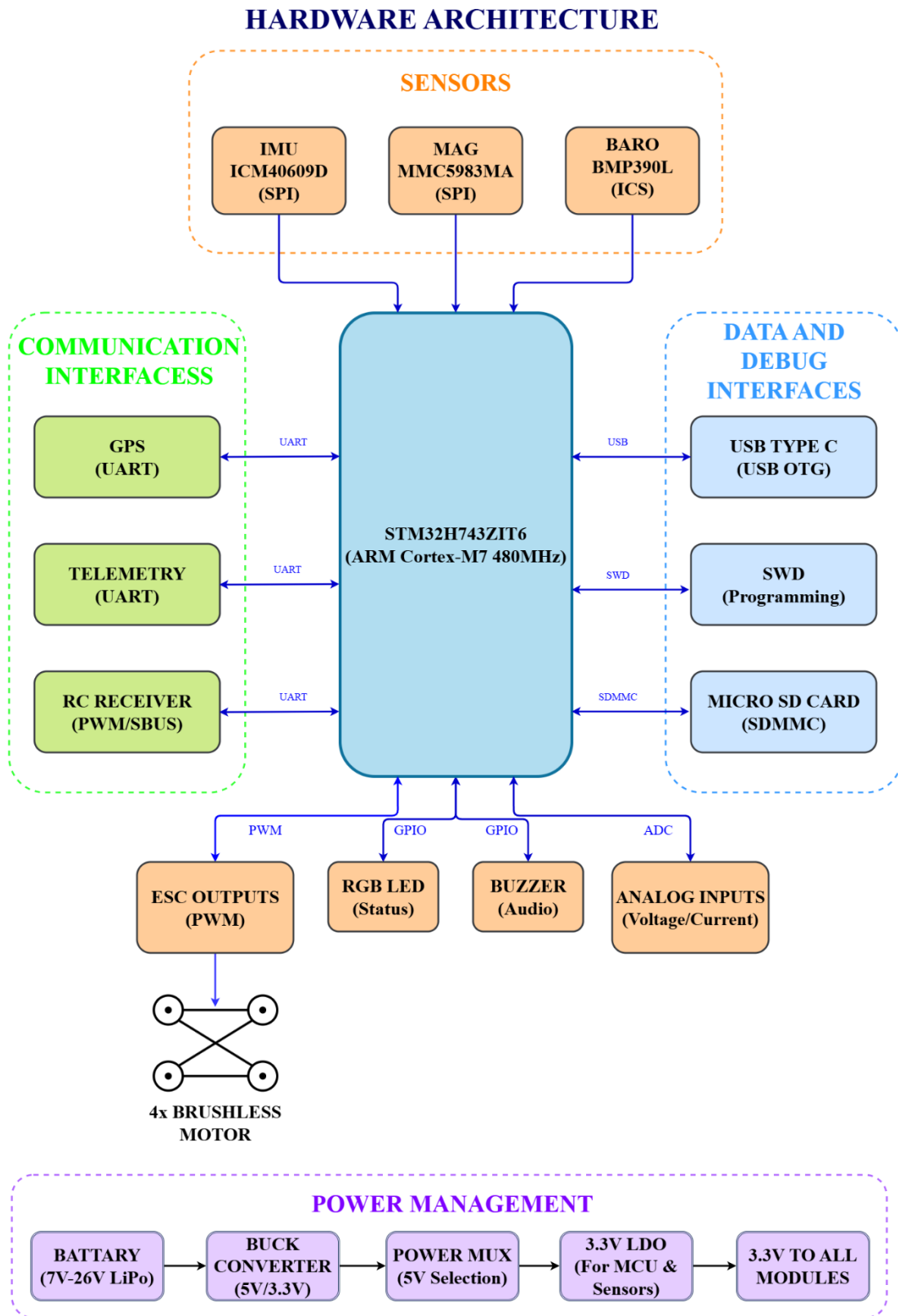


Figure 3.1 -- Hardware Design Flow

3.2 Design Methodology

The flight controller hardware was developed in five major stages.

Stage 1 -- Component Selection

The first stage involved selecting suitable components based on performance, availability, compatibility, and future expandability. The main hardware components used are listed in Table 3.1.

Component	Function
STM32CUBEMX	Main Microcontroller
ICM-40609D	Inertial Measurement Unit
MMC5983MA	Magnetometer
BMP390L	Barometric Pressure Sensor
USB Type-C Connector	Programming & Communication
Buck Converter	Voltage Conversion
Power Multiplexer	Power Source Selection
3.3V LDO	Voltage Regulation
Micro SD Card Slot	Data Logging
SWD Header	Programming & Debugging
RGB LED	Status Indication
Buzzer	Audible Notification
ESC Connectors	Motor Outputs
GPS Connector	GPS Interface
Telemetry Connector	Telemetry Interface
RC Receiver Connector	Receiver Interface

Table 3.1 -- Hardware Components Used in Schematic Design

Stage 2 -- Schematic Design

The complete schematic was designed using Proteus Professional. To improve readability, the design was divided into four functional sheets:

- Main Controller Sheet
- Sensor Sheet
- Power Supply Sheet

- This modular approach simplifies future maintenance and debugging.

The first schematic sheet contains the STM32CUBEMX microcontroller and its supporting circuitry. The circuit includes:

- STM32CUBEMX microcontroller
- Clock oscillator
- Reset circuit
- Boot configuration
- Decoupling capacitors
- Power pins
- SWD programming interface
- Communication pin breakout

The schematic illustrates the internal circuitry of a flight controller PCB. Key components include:

- Microcontrollers:** U1A (STM32H7432T6), U1B (STM32H7432T6), U1C (STM32H7432T6), and U1D (STM32H7432T6).
- Op-Amp:** U2 (PC14060C2, PC14060C2).
- Logic Gate:** U3 (C7E180004107G13).
- Power Management:** VREF FILTER (C17, C18), CRYSTAL OSCILLATOR (C19), and RGB LED (C21).
- Other Components:** C16, C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, D1, D2, D3, D4, D5, D6, D7, D8, D9, D10, D11, D12, D13, D14, D15, D16, D17, D18, D19, D20, D21, D22, D23, D24, D25, D26, D27, D28, D29, D30, D31, D32, D33, D34, D35, D36, D37, D38, D39, D40, D41, D42, D43, D44, D45, D46, D47, D48, D49, D50, D51, D52, D53, D54, D55, D56, D57, D58, D59, D60, D61, D62, D63, D64, D65, D66, D67, D68, D69, D70, D71, D72, D73, D74, D75, D76, D77, D78, D79, D80, D81, D82, D83, D84, D85, D86, D87, D88, D89, D90, D91, D92, D93, D94, D95, D96, D97, D98, D99, D100, D101, D102, D103, D104, D105, D106, D107, D108, D109, D110, D111, D112, D113, D114, D115, D116, D117, D118, D119, D120, D121, D122, D123, D124, D125, D126, D127, D128, D129, D130, D131, D132, D133, D134, D135, D136, D137, D138, D139, D140, D141, D142, D143, D144, D145, D146, D147, D148, D149, D150, D151, D152, D153, D154, D155, D156, D157, D158, D159, D160, D161, D162, D163, D164, D165, D166, D167, D168, D169, D170, D171, D172, D173, D174, D175, D176, D177, D178, D179, D180, D181, D182, D183, D184, D185, D186, D187, D188, D189, D190, D191, D192, D193, D194, D195, D196, D197, D198, D199, D200, D201, D202, D203, D204, D205, D206, D207, D208, D209, D210, D211, D212, D213, D214, D215, D216, D217, D218, D219, D220, D221, D222, D223, D224, D225, D226, D227, D228, D229, D230, D231, D232, D233, D234, D235, D236, D237, D238, D239, D240, D241, D242, D243, D244, D245, D246, D247, D248, D249, D250, D251, D252, D253, D254, D255, D256, D257, D258, D259, D260, D261, D262, D263, D264, D265, D266, D267, D268, D269, D270, D271, D272, D273, D274, D275, D276, D277, D278, D279, D280, D281, D282, D283, D284, D285, D286, D287, D288, D289, D290, D291, D292, D293, D294, D295, D296, D297, D298, D299, D300, D301, D302, D303, D304, D305, D306, D307, D308, D309, D310, D311, D312, D313, D314, D315, D316, D317, D318, D319, D320, D321, D322, D323, D324, D325, D326, D327, D328, D329, D330, D331, D332, D333, D334, D335, D336, D337, D338, D339, D340, D341, D342, D343, D344, D345, D346, D347, D348, D349, D350, D351, D352, D353, D354, D355, D356, D357, D358, D359, D360, D361, D362, D363, D364, D365, D366, D367, D368, D369, D370, D371, D372, D373, D374, D375, D376, D377, D378, D379, D380, D381, D382, D383, D384, D385, D386, D387, D388, D389, D390, D391, D392, D393, D394, D395, D396, D397, D398, D399, D400, D401, D402, D403, D404, D405, D406, D407, D408, D409, D410, D411, D412, D413, D414, D415, D416, D417, D418, D419, D420, D421, D422, D423, D424, D425, D426, D427, D428, D429, D430, D431, D432, D433, D434, D435, D436, D437, D438, D439, D440, D441, D442, D443, D444, D445, D446, D447, D448, D449, D450, D451, D452, D453, D454, D455, D456, D457, D458, D459, D460, D461, D462, D463, D464, D465, D466, D467, D468, D469, D470, D471, D472, D473, D474, D475, D476, D477, D478, D479, D480, D481, D482, D483, D484, D485, D486, D487, D488, D489, D490, D491, D492, D493, D494, D495, D496, D497, D498, D499, D500, D501, D502, D503, D504, D505, D506, D507, D508, D509, D510, D511, D512, D513, D514, D515, D516, D517, D518, D519, D520, D521, D522, D523, D524, D525, D526, D527, D528, D529, D530, D531, D532, D533, D534, D535, D536, D537, D538, D539, D540, D541, D542, D543, D544, D545, D546, D547, D548, D549, D550, D551, D552, D553, D554, D555, D556, D557, D558, D559, D560, D561, D562, D563, D564, D565, D566, D567, D568, D569, D570, D571, D572, D573, D574, D575, D576, D577, D578, D579, D580, D581, D582, D583, D584, D585, D586, D587, D588, D589, D590, D591, D592, D593, D594, D595, D596, D597, D598, D599, D600, D601, D602, D603, D604, D605, D606, D607, D608, D609, D610, D611, D612, D613, D614, D615, D616, D617, D618, D619, D620, D621, D622, D623, D624, D625, D626, D627, D628, D629, D630, D631, D632, D633, D634, D635, D636, D637, D638, D639, D640, D641, D642, D643, D644, D645, D646, D647, D648, D649, D650, D651, D652, D653, D654, D655, D656, D657, D658, D659, D660, D661, D662, D663, D664, D665, D666, D667, D668, D669, D670, D671, D672, D673, D674, D675, D676, D677, D678, D679, D680, D681, D682, D683, D684, D685, D686, D687, D688, D689, D690, D691, D692, D693, D694, D695, D696, D697, D698, D699, D700, D701, D702, D703, D704, D705, D706, D707, D708, D709, D710, D711, D712, D713, D714, D715, D716, D717, D718, D719, D720, D721, D722, D723, D724, D725, D726, D727, D728, D729, D730, D731, D732, D733, D734, D735, D736, D737, D738, D739, D740, D741, D742, D743, D744, D745, D746, D747, D748, D749, D750, D751, D752, D753, D754, D755, D

Figure 3.2 -- Main Controller Schematic

3.4 Sensor Interface Circuit

The second schematic sheet contains all onboard sensors. The sensors are connected to the STM32CUBEMX through dedicated communication interfaces.

ICM40609D IMU

The IMU provides three-axis acceleration and three-axis angular velocity measurements. Communication Protocol: SPI. Purpose: Motion sensing, orientation measurement.

MMC5983MA Magnetometer

The magnetometer measures the Earth's magnetic field to determine heading. Communication Protocol: SPI. Purpose: Compass functionality, direction estimation.

BMP390L Barometer

The BMP390L measures atmospheric pressure and temperature. Communication Protocol: I²C. Purpose: Altitude estimation, vertical positioning.

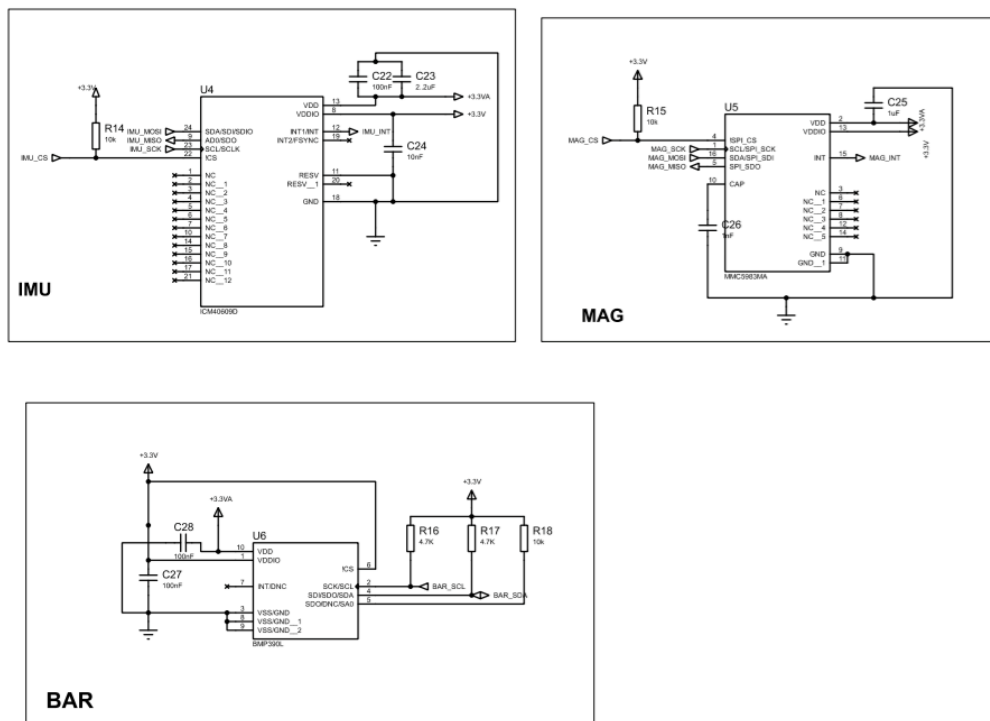


Figure 3.3 -- Sensor Interface Schematic

FILE NAME: FINAL DRONE.pdssprj	DATE: 09-07-202
DESIGN TITLE: FLIGHT CONTROLLER	FILE: FINAL DRONE.pdssprj
PATH: D:\DRONE\Final Drone\MINI_PROJECT_FINAL\	2 of 4
BY: BOARD:@BOARD	REV: TIME: 00:02

3.5 Power Management Circuit

The third schematic sheet implements the power management section of the flight controller. The power circuit includes:

- Input power connector
- Buck converter
- Power multiplexer
- 3.3V Low Dropout Regulator
- Power filtering capacitors
- Protection circuitry

The buck converter reduces the input voltage to a suitable intermediate level, while the LDO generates a stable 3.3V supply for the microcontroller and sensors. Special attention was given to voltage stability, ripple reduction, noise filtering, and proper decoupling.

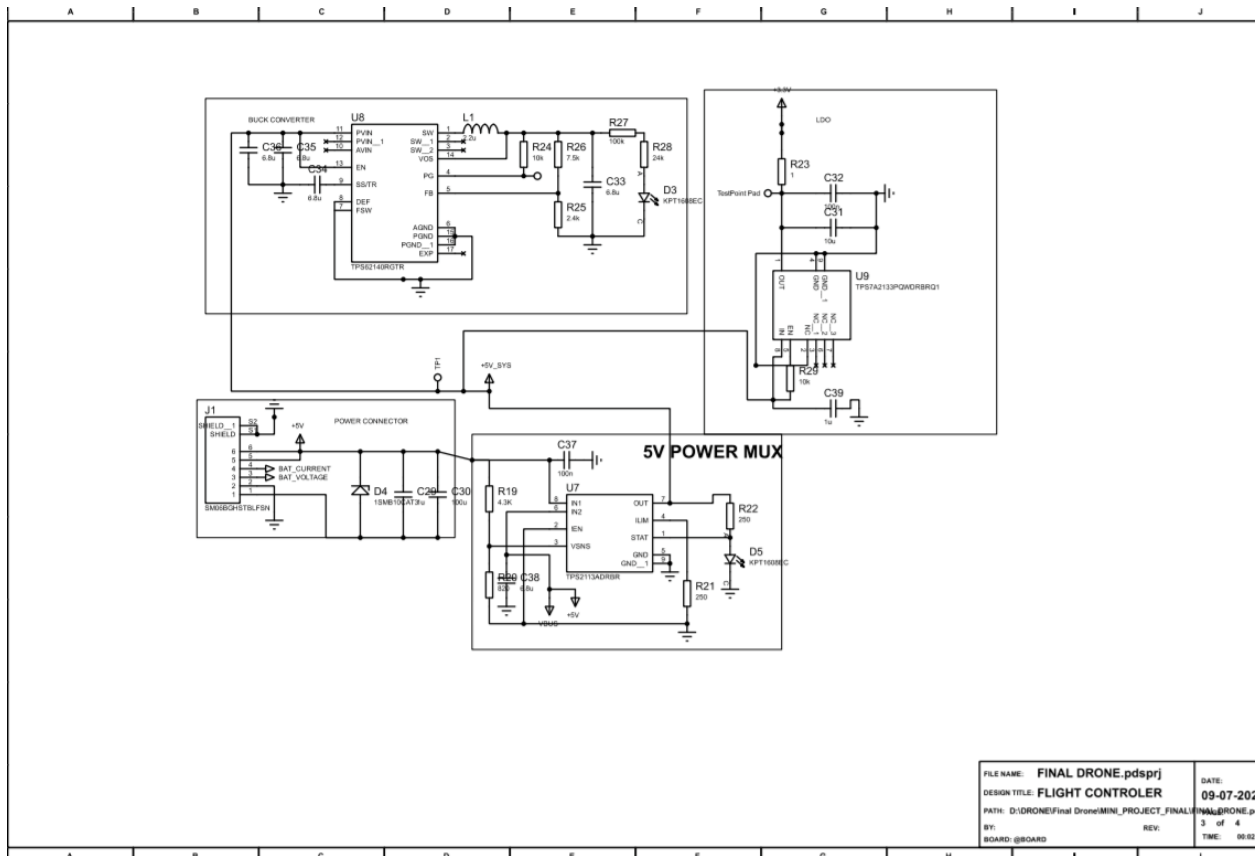


Figure 3.4 -- Power Management Schematic

3.6 USB and External Interface Circuit

The fourth schematic sheet contains the external communication interfaces. These include:

- USB Type-C
- GPS Connector
- Telemetry Connector
- RC Receiver Connector
- SD Card Interface
- ESC Output Connectors
- RGB LED
- Buzzer

These interfaces provide flexibility for future expansion and integration with other UAV subsystems.

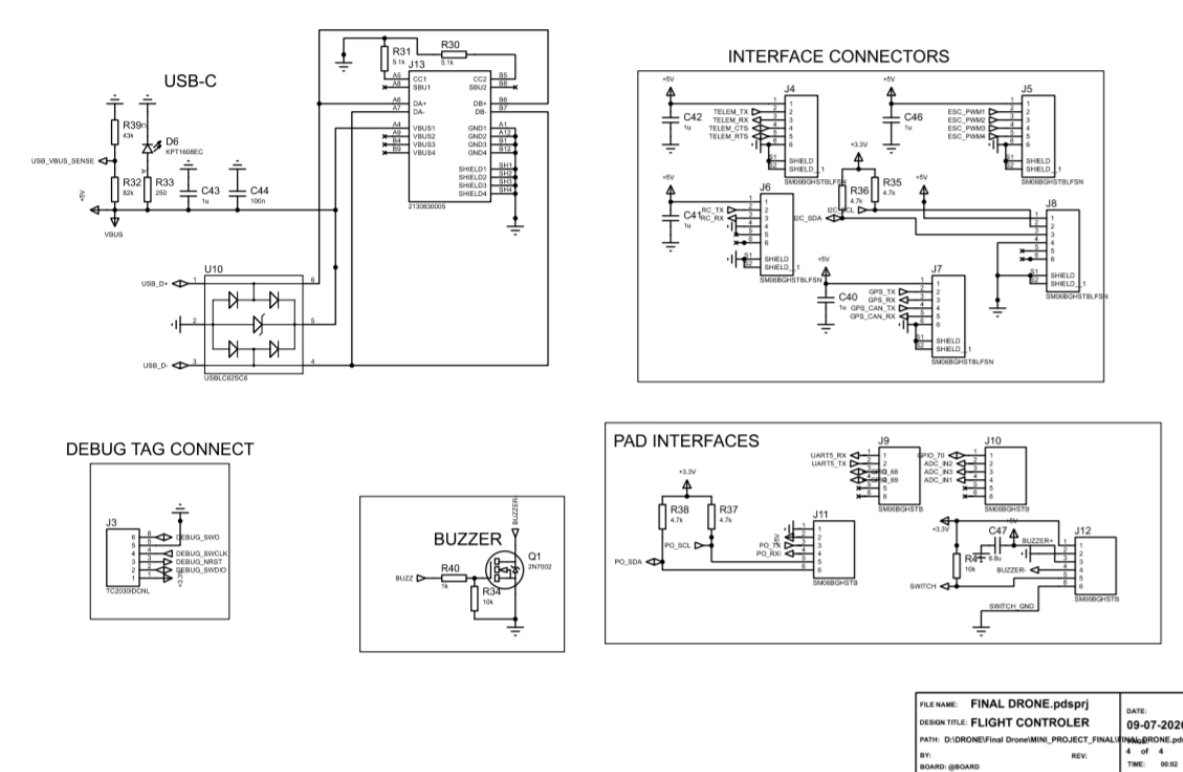


Figure 3.5 -- USB and Interface Schematic

3.7 STM32CUBEMX Pin Configuration

After completing the schematic, the STM32CUBEMX was configured using STM32CUBEMX. The pin configuration was prepared according to the requirements of the hardware design. The following peripherals were enabled:

- GPIO

- SPI
- I²C
- UART
- USB OTG
- SDMMC
- ADC
- PWM Timers
- SWD Debug

Each peripheral was assigned appropriate pins based on the schematic connections to ensure compatibility between the software configuration and hardware design.

Peripheral	Purpose
GPIO	General Input/Output
SPI	IMU and Magnetometer Communication
I ² C	BMP390L Communication
UART	GPS and Telemetry
USB OTG	USB Type-C Interface
SDMMC	SD Card Interface
ADC	Analog Inputs
PWM Timers	ESC Outputs
SWD	Programming and Debugging

Table 3.2 -- STM32 Peripheral Configuration

- Continuous ground reference
- Adequate spacing between components
- Accessible connectors

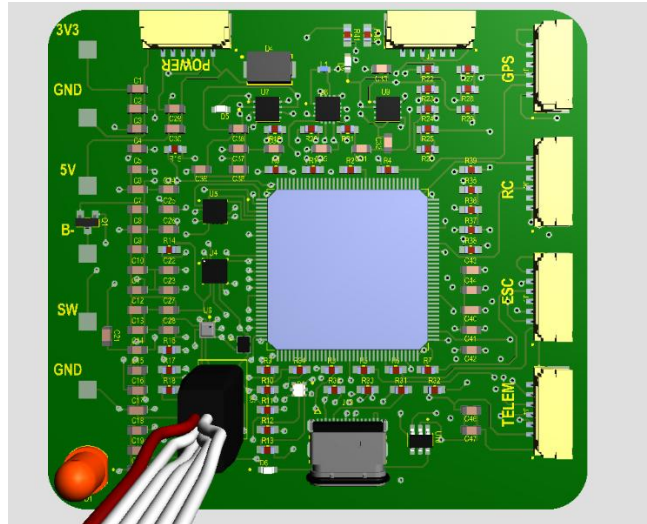


Figure 3.7 -- PCB 3D Top Layer

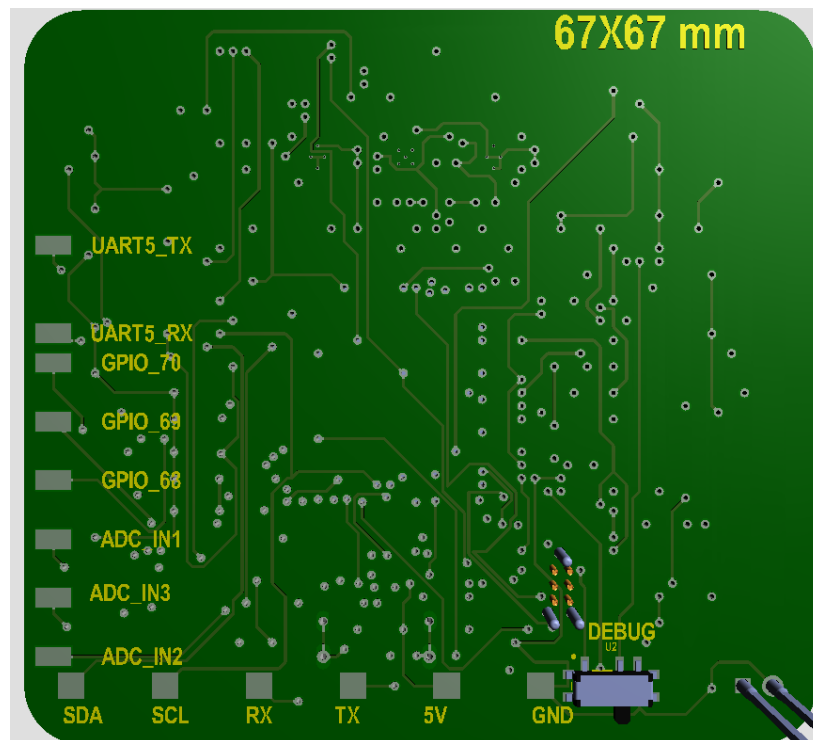


Figure 3.8 – PCB 3D Bottom Layer

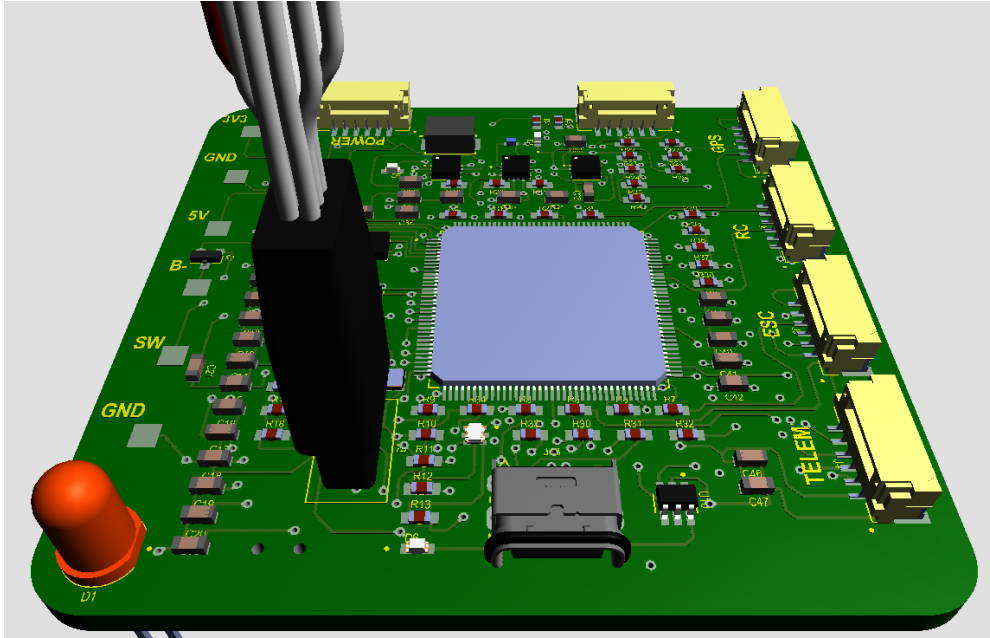


Figure 3.9 -- 3D PCB View

3.10 PCB Design Considerations

The following design practices were followed during PCB development:

- Decoupling capacitors were placed close to the power pins of the STM32CUBEMX
- High-speed communication traces were kept as short as possible
- Power traces were made wider than signal traces to reduce voltage drop
- The crystal oscillator was positioned close to the microcontroller
- A continuous ground plane was used to reduce electromagnetic interference
- Connectors were placed at the PCB edges to simplify external wiring
- Design Rule Check (DRC) was performed after routing to identify and eliminate design violations

CHAPTER 4

PCB DESIGN VERIFICATION AND ANALYSIS

4.1 Introduction

After completing the schematic design and PCB layout, it is essential to verify the design to ensure that all electrical connections are correct and that the PCB complies with standard design rules. Verification helps identify potential errors before fabrication, reducing the possibility of hardware failure and minimising redesign efforts.

In this project, the verification process consisted of checking the schematic connections, reviewing the PCB layout, confirming the STM32CUBEMX pin assignments, and ensuring that the board was ready for future fabrication and firmware development. The design was analysed with respect to power distribution, component placement, signal routing, and communication interfaces.

4.2 STM32CUBEMX Pin Configuration Verification

The STM32CUBEMX was configured using STM32CUBEMX to match the hardware connections defined in the schematic. The following peripheral configurations were reviewed:

- SPI interfaces for sensor communication
- I²C interface for the BMP390L
- UART interfaces for GPS and telemetry
- USB OTG interface for USB Type-C
- SDMMC interface for the microSD card
- PWM timer outputs for ESC connectors
- GPIO pins for RGB LED, buzzer, and other control signals
- SWD interface for programming and debugging

The software pin configuration was checked against the schematic to ensure consistency between the hardware design and future firmware development.

4.3 PCB Layout Analysis

The PCB layout was designed to achieve a compact, organised, and reliable hardware platform. During layout development, emphasis was placed on minimising routing complexity while maintaining proper spacing between components.

The microcontroller was positioned near the centre of the PCB to reduce the average routing distance to peripheral components. Sensors were grouped together to simplify communication routing, and the power management section was separated from the sensor area to reduce electrical noise.

External connectors such as USB Type-C, GPS, telemetry, RC receiver, and ESC outputs were placed near the board edges to facilitate easy access during assembly and future integration.

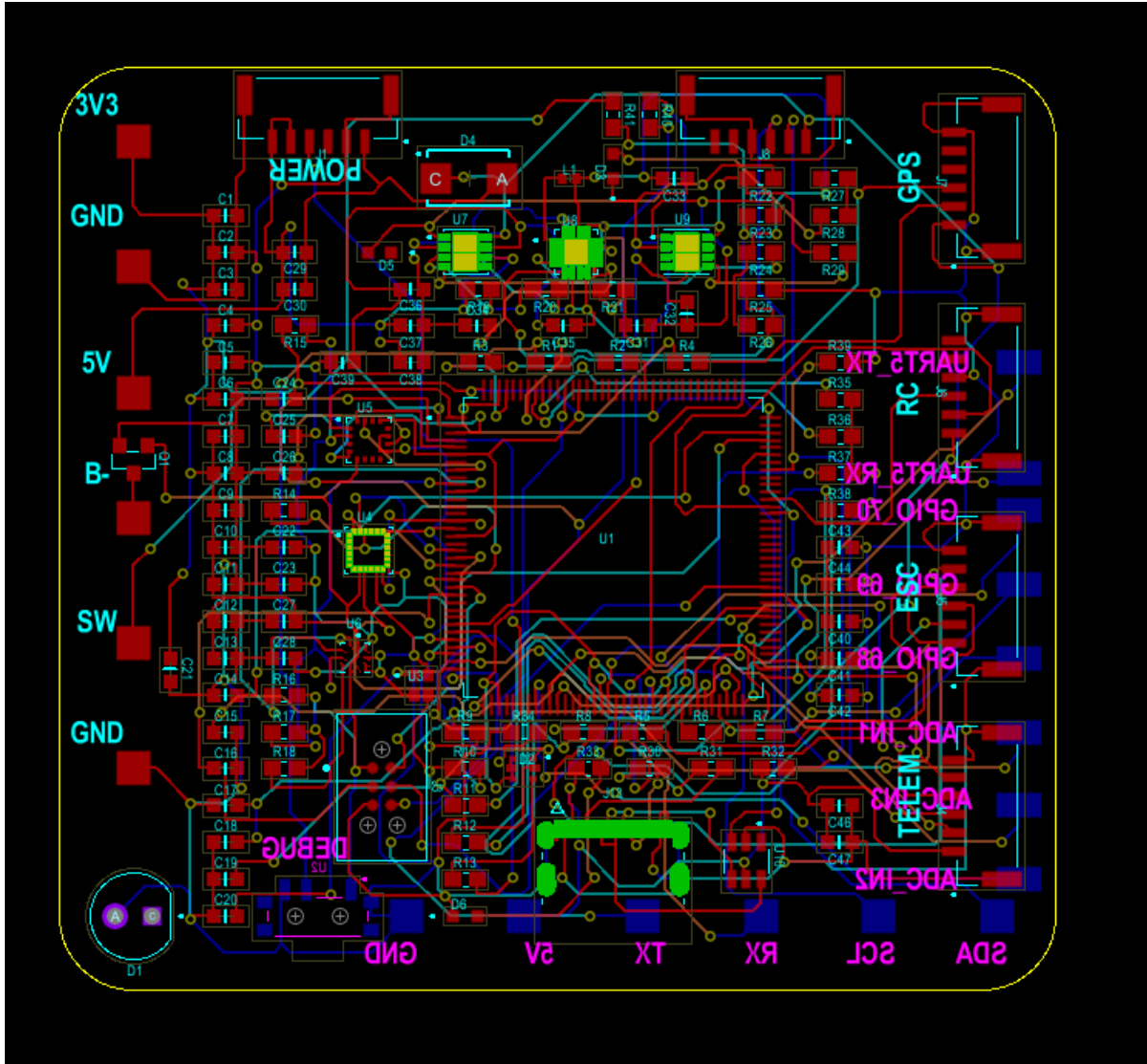


Figure 4.0 -- PCB Layout

4.4 PCB Routing Analysis

The routing process was carried out by considering standard PCB design practices. The following design considerations were adopted:

- Power traces were made wider than signal traces to improve current-carrying capability
- High-speed communication traces were kept as short as possible
- The crystal oscillator was placed close to the STM32CUBEMX
- Signal routing was arranged to avoid unnecessary crossings
- Sensitive sensor connections were kept away from the power section wherever possible

- Connectors were positioned to simplify external wiring

These routing practices contribute to better signal integrity and improve the overall reliability of the PCB.

4.5 Design Rule Verification

Before considering the PCB design complete, the layout should be checked using the Design Rule Check (DRC) feature available in Proteus. The DRC verifies parameters such as:

- Minimum track width
- Clearance between tracks
- Pad spacing
- Via spacing
- Unconnected nets
- Short circuits
- Copper overlaps

Performing these checks ensures that the PCB meets the selected manufacturing rules and reduces the likelihood of fabrication errors.

```

Pre-production check start.
File: D:/DRONE/Final Drone/MINI_PROJECT_FINAL/FINAL DRONE.pdsprj
Date: 15 July 2026, 20:08:37

TEST: Board edge
PASS: Board edge complete.

TEST: Components board placement
PASS: Components within board edge.

TEST: Differential pairs
PASS: Differential pairs compliant.

TEST: DRC
PASS: No DRC errors.

TEST: DRC room rules assigned
PASS: DRC room and rules tally.

TEST: Duplicate part IDs
PASS: All part IDs are unique.

TEST: Layer stackup and drill sets
PASS: Layer stackup valid.

TEST: Length matched routes
PASS: Length matched routes within tolerances.

TEST: Managed library parts
PASS: Managed library mode is disabled.

TEST: Managed library styles
PASS: Managed library mode is disabled.

TEST: Net none routing
PASS: No net none routing found.

PASS: No overlap detected.

TEST: Object validity
PASS: Objects valid

TEST: Pad-stack validity
PASS: No invalid pad-stacks in use.

TEST: Stitching-vias connectivity
PASS: Stitching vias stitch correctly.

TEST: Test points
PASS: Test points valid.

TEST: Trace validity
PASS: Traces valid.

TEST: Via and drill range overlaps
PASS: No via or drill range overlaps found.

TEST: Via layer spans
PASS: Vias valid.

TEST: Zone hatched connections
PASS: Hatched zone check complete.

TEST: Zone presence
PASS: No voided zones found.

Pre-production check end:
0 errors, 0 warnings, 2 failed, 0 skipped, 21 passed.

```

Figure 4.1 -- Design Rule Check (DRC)

4.6 Electrical Rule Verification

The Electrical Rule Check (ERC) is used to verify the logical correctness of the schematic. The ERC helps identify issues such as:

- Missing power connections
- Floating input pins

- Incorrect net assignments
- Conflicting signal directions
- Unconnected components

After resolving any reported issues, the schematic is considered ready for PCB generation.

```
#I:ISIS Release 9.00.02 (Build 40482) (C) Labcenter Electronics 1990- 2025.
#I:Compiling design 'D:/DRONE/Final Drone/MINI_PROJECT_FINAL/FINAL DRONE.pdsprj'.
%C=0002,00000006

ELECTRICAL RULES CHECK
=====
Design:    FLIGHT CONTROLLER
Doc. no.:  1
Revision:
Author:
Created:   15-05-2026
Modified:  15-07-2026

UNDRIVEN: U1:A,VSS__1 (Power Pin)
UNDRIVEN: U1:A,VSS__2 (Power Pin)
UNDRIVEN: U1:A,VSS__3 (Power Pin)
UNDRIVEN: U1:A,VSS__4 (Power Pin)
UNDRIVEN: U1:A,VSS__5 (Power Pin)
UNDRIVEN: U1:A,VSS__6 (Power Pin)
UNDRIVEN: U1:A,VSS__7 (Power Pin)
UNDRIVEN: U1:A,VSS__8 (Power Pin)
UNDRIVEN: U1:A,PDR_ON (Input)
UNDRIVEN: U8,PVIN__1 (Input)
UNDRIVEN: U8,AVIN (Input)
UNDRIVEN: U8,EXP (Power Pin)
UNDRIVEN: U8,FSW (Input)
UNDRIVEN: J13,VBUS4 (Power Pin)
UNDRIVEN: J13,VBUS3 (Power Pin)
UNDRIVEN: J13,VBUS2 (Power Pin)
SINGULAR LABEL: PO_RX (Input)
WARNING: PO_RX,<TERM> (Input) connected to U1:A,PA3 (I/O)
SINGULAR LABEL: SD_DAT0 (I/O)
SINGULAR LABEL: SD_DAT1 (I/O)
SINGULAR LABEL: SD_DAT2 (I/O)
SINGULAR LABEL: SD_DAT3 (I/O)
SINGULAR LABEL: SD_CLK (Output)
SINGULAR LABEL: GPS_CANRX (Input)
WARNING: GPS_CANRX,<TERM> (Input) connected to U1:C,PD0 (I/O)
SINGULAR LABEL: SD_CMD (Output)
SINGULAR LABEL: SD_CDS (Input)
WARNING: SD_CDS,<TERM> (Input) connected to U1:C,PD3 (I/O)
SINGULAR LABEL: PWR_GOOD (Input)
WARNING: PWR_GOOD,<TERM> (Input) connected to U1:C,PD9 (I/O)
SINGULAR LABEL: BAT_VOLTAGE (Input)
WARNING: BAT_VOLTAGE,<TERM> (Input) connected to U1:D,PF11 (I/O)
WARNING: U4,INT1/INT (Output) connected to U1:D,PF5 (I/O)
WARNING: U5,SPI_SDO (Output) connected to U1:C,PE5 (I/O)
WARNING: U5,INT (Output) connected to U1:B,PC14/OSC32_IN (I/O)
SINGULAR LABEL: TESTPOINT PAD (Generic)
SINGULAR LABEL: BAT_VOLTAGE (Output)
SINGULAR LABEL: PO_RX\ (Output)
UNDRIVEN: PO_RX\,<TERM> (Output)
SINGULAR LABEL: GPS_CAN_RX (Output)
SINGULAR LABEL: BUZZER+ (LBL)
SINGULAR LABEL: TP1 (Generic)
ERROR:   +5V_SYS,<TERM> (Power Rail) connected to U7,OUT (Output)
SINGULAR LABEL: SWITCH_GND (LBL)

Netlist generated OK.
ERC errors found.
```

Figure 4.2 -- Electrical Rule Check (ERC)

4.7 Advantages of the Proposed Hardware Design

The custom flight controller developed in this project offers several advantages.

Advantage	Description
High-Performance MCU	Uses STM32CUBEMX ARM Cortex-M7 microcontroller.
Modular Design	Schematic divided into functional sections for easier understanding and maintenance.
Expandable Hardware	Includes interfaces for GPS, telemetry, SD card, RC receiver, and ESCs.
Professional PCB Layout	Designed using standard PCB layout practices.
Modern Connectivity	USB Type-C interface and SWD debugging support.
Essential Sensors	Integrates IMU, magnetometer, and barometer.
Ready for Future Development	Hardware prepared for firmware implementation and drone integration.

Table 4.1 -- Advantages of the Proposed Design

4.8 Limitations of the Current Work

The current project focuses only on the hardware design stage. The following activities have not been completed and are outside the scope of this mini project:

- PCB fabrication
- Component assembly and soldering
- Firmware development
- Sensor calibration
- Hardware testing
- Motor control testing
- Flight testing
- Autonomous navigation implementation

These limitations are expected because the objective of the project was to design the hardware platform rather than develop a complete flight-ready system.

4.9 Discussion

The completed schematic, PCB layout, and STM32CUBEMX pin configuration provide a solid foundation for future development of a custom UAV flight controller. By using the STM32CUBEMX microcontroller and integrating essential sensors and interfaces, the design supports future implementation of flight control firmware and expansion with additional modules.

The modular organisation of the schematic simplifies future modifications, while the PCB layout follows good engineering practices that will assist during fabrication and assembly. Overall, the project demonstrates the application of embedded hardware design principles in developing a professional flight controller suitable for educational and research purposes.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The objective of this mini project was to design a custom flight controller hardware for UAV applications using the STM32CUBEMX microcontroller. The project focused exclusively on the hardware development stage, including schematic design, PCB layout, and STM32CUBEMX pin configuration. These are essential steps in developing a reliable flight controller and provide the foundation for future firmware implementation and drone development.

A complete schematic of the flight controller was successfully designed using Proteus Design Suite. The schematic integrates the STM32CUBEMX microcontroller with the essential hardware required for a modern flight controller, including the ICM40609D Inertial Measurement Unit (IMU), MMC5983MA magnetometer, BMP390L barometric pressure sensor, USB Type-C interface, SWD debugging interface, SD card interface, GPS connector, telemetry connector, RC receiver connector, ESC outputs, RGB status LED, buzzer, and power management circuitry. The modular organisation of the schematic into separate functional sections improves readability, simplifies future modifications, and facilitates debugging.

Following schematic completion, a PCB layout was developed in Proteus. During the PCB design process, attention was given to component placement, routing strategy, power distribution, and signal integrity. Components were arranged to reduce routing complexity and improve accessibility, while high-speed signals and power traces were routed following standard embedded hardware design practices. The PCB layout is suitable for future fabrication and assembly.

The STM32CUBEMX was configured using STM32CUBEMX, where the required peripherals were assigned according to the hardware design. Interfaces such as SPI, I²C, UART, USB OTG, SDMMC, PWM, ADC, GPIO, and SWD were configured to support communication with the onboard sensors and external peripherals. This ensures consistency between the hardware design and future software development.

Although firmware development, PCB fabrication, hardware testing, and flight experiments were not included in the scope of this project, the completed design represents a strong foundation for future implementation. The hardware is modular, expandable, and suitable for integration into a custom UAV platform.

This project has provided practical experience in embedded hardware design, schematic capture, PCB layout development, component selection, and microcontroller configuration. It has also enhanced understanding of the hardware architecture of modern flight controllers and the design practices required for developing reliable embedded systems.

Overall, the objectives of the mini project have been successfully achieved, and the resulting design is ready for the next stages of development.

5.2 Future Scope

The current work is limited to the hardware design of the flight controller. Several stages remain to transform the design into a fully functional UAV flight control system. The proposed future work includes the following.

5.2.1 PCB Fabrication

The designed PCB can be manufactured using professional PCB fabrication services. After fabrication, all electronic components can be assembled according to the schematic.

5.2.2 Hardware Assembly and Testing

After PCB fabrication, the hardware can be assembled and tested. Basic verification such as power supply validation, clock generation, communication interface testing, and sensor connectivity can be performed to ensure correct operation.

5.2.3 Firmware Development

The next stage will involve developing firmware using STM32CUBEMX. The firmware will initialise all peripherals, configure communication interfaces, read sensor data, and control the outputs required for flight operation.

5.2.4 Sensor Calibration

The onboard sensors, including the IMU, magnetometer, and barometer, can be calibrated to improve measurement accuracy. Proper calibration is essential for reliable attitude estimation and stable flight.

5.2.5 Development of a Complete Quadcopter

The custom flight controller developed in this project can be integrated into a complete quadcopter platform. After firmware development and testing, the controller can be connected with Electronic Speed Controllers (ESCs), brushless motors, a power distribution system, and other required hardware to create a fully functional UAV.

5.2.6 Raspberry Pi Integration for Autonomous Flight

A Raspberry Pi can be integrated as a companion computer to provide advanced processing capabilities that complement the STM32CUBEMX flight controller. In the proposed architecture, the STM32CUBEMX will perform real-time flight control, sensor acquisition, and motor control, while the Raspberry Pi will execute computationally intensive tasks such as image processing, computer vision, path planning, and decision-making. This combination enables the development of an autonomous UAV capable of performing advanced missions while maintaining reliable real-time flight control.

5.2.7 Autonomous Navigation

After integrating the Raspberry Pi, autonomous capabilities can be implemented, including:

- Waypoint navigation
- Obstacle detection and avoidance
- Vision-based navigation
- Object detection and tracking
- Autonomous mission execution

These features will allow the drone to perform complex tasks with minimal human intervention.

5.3 Overall Achievement of the Project

The major achievements of this mini project are summarised in Table 5.1.

Objective	Status
Literature Review	Completed
Hardware Component Selection	Completed
Flight Controller Architecture Design	Completed
Proteus Schematic Design	Completed
STM32CUBEMX Circuit Design	Completed
Sensor Interface Design	Completed
Power Management Circuit Design	Completed
STM32CUBEMX Pin Configuration	Completed
PCB Layout Design	Completed
PCB Fabrication	Future Work
Hardware Testing	Future Work
Firmware Development	Future Work
Complete Drone Development	Future Work
Raspberry Pi Integration	Future Work
Autonomous Flight	Future Work

Table 5.1 -- Project Achievement Summary

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